MIL-F-16377/2A(SH)
29 July 1981
SUPERSEDING
MIL-F-16377/2(SHIPS)
30 November 1970

MILITARY SPECIFICATION SHEET

(A) FIXTURES, LIGHTING; FLUORESCENT, DETAIL LIGHTING FOR WATER GAGES

6 WATTS, 120 VOLTS, 60 HERTZ, SYMBOLS 152 AND 154

This specification sheet is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the fixtures described herein shall consist of this document and the latest issue of MIL-F-16377(SH).

(A)

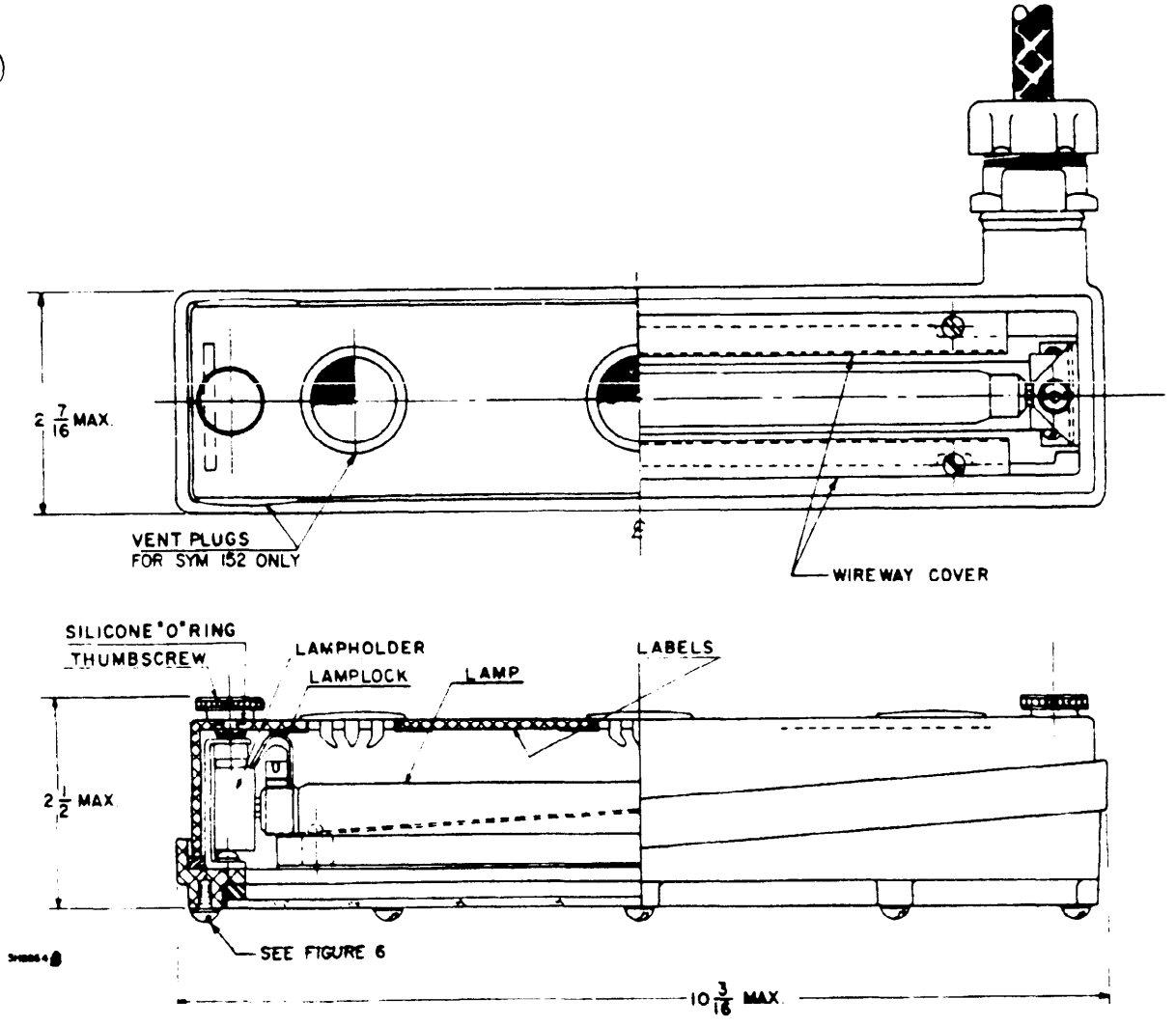


FIGURE 1. Typical view of Symbols 152 and 154.

All dimensions in inches.
For tolerances, other than specified herein, see paragraph 3.5.2 of MIL-F-16377.

(A) Denotes changes

FSC 6210

Source: https://assist.dla.mil -- Downloaded: 20261O-18T08:39Z
Check the source to verify that this is the current version before use.

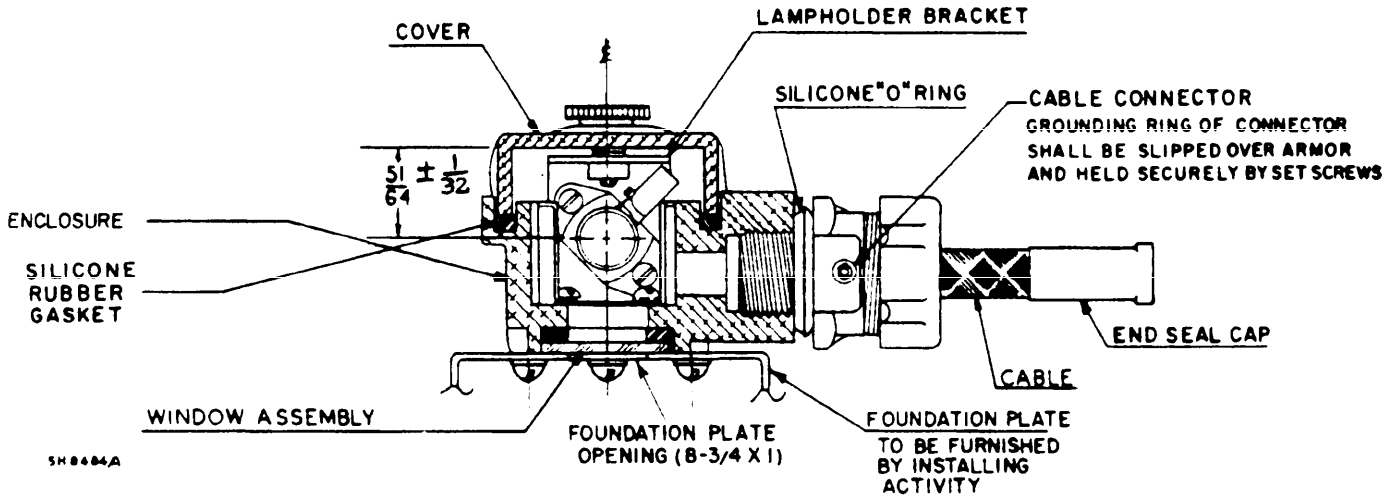
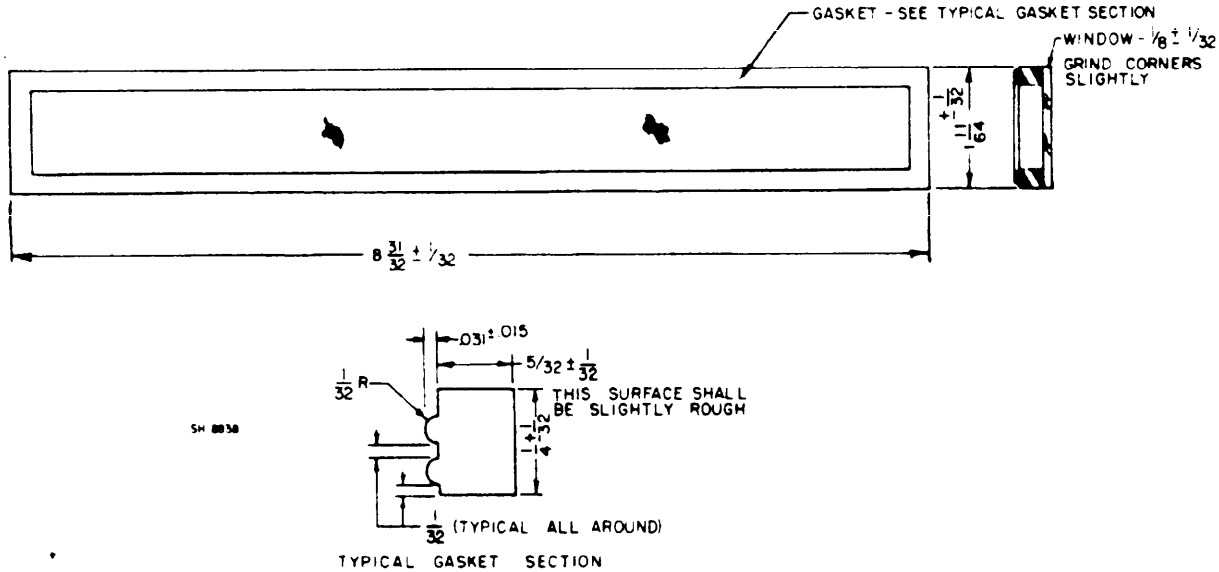


FIGURE 2. Typical section.

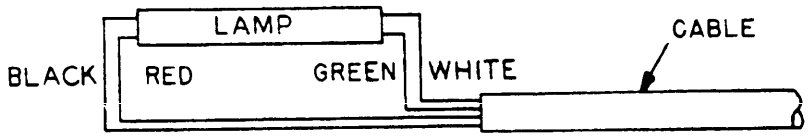


TYPICAL GASKET SECTION

NOTE: DIMENSIONS SHOWN ARE FOR GLASS PORTION OF ASSEMBLY. GASKET SHALL BE FLUSH WITH OR MAY PROJECT SLIGHTLY OVER LENGTH AND WIDTH OF GLASS.

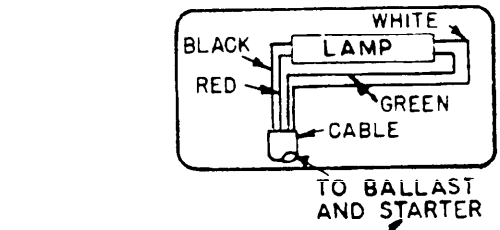
FIGURE 3. Window assembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:39Z
Check the source to verify that this is the current version before use.



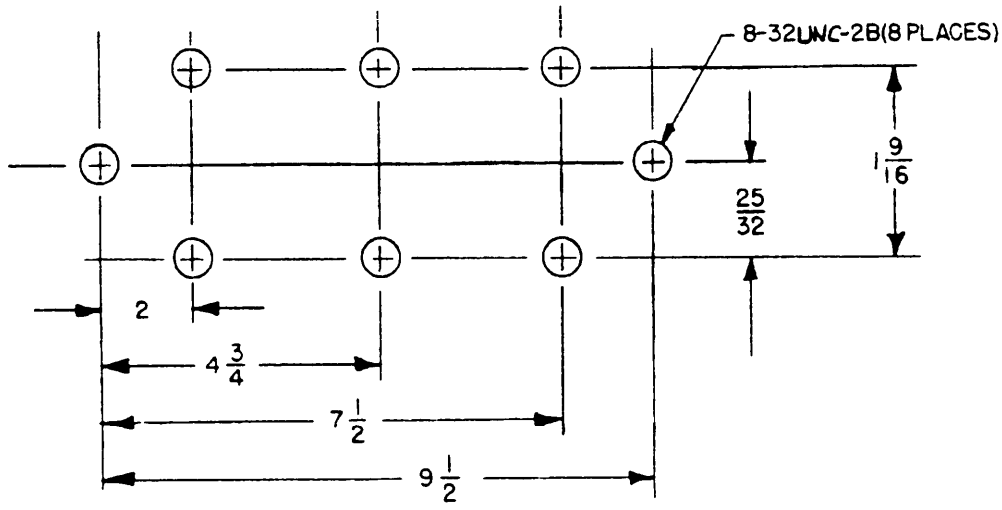
SH 8485

FIGURE 4. Wiring diagram.



SH 8486

FIGURE 5. Type III label.



SH 9991

FIGURE 6. Mounting dimensions of bottom of enclosure.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:39Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

CLASSIFICATION:
Type: I
Class: 1

SYMBOL NUMBERS:
152: Light, ventilated, for use with one 6-watt cool white lamp.
154: Light, watertight, for use with one 6-watt cool white lamp.

DESIGN:
(A) Electrical rating: 6 watts, 120 volts, 60 Hz.
Dimensions and configuration: See figure 1.
Typical section: See figure 2.
Operating temperature: 275°F. maximum.
Cable: 10 feet, type FSGU-3of MIL-C-915/31. The free end of the cable shall be end sealed to prevent entrance of moisture (see figure 2).
Enclosure:
Symbol 152: Totally enclosed.
Symbol 154: Watertight.
Housing: Cast aluminum
Window assembly: See figure 3. Window glass shall be clear transparent, heat resistant, high impact. The window gasket shall be silicone rubber, class 2b , grade 50, in accordance with ZZ-R-765 and shall be attached to the window with cement.
Wiring diagram: See figure 4.
Mounting dimensions: See figure 6.
Mounting plate: To be designed by the manufacturer to suit the particular gage with which the light is fitted, as specified in the contract or order.
Lamp: W-L-116/1; Commercial designation F6T5/CW, NSN 6240-00-299-7252).
Lampholder: MIL-L-970/15, (NSN 6250-00-075-5666).
Lamp lock: M16377/40-001.
Designation and marking: The following labels shall be furnished in accordance with MIL-F-16377 applied as shown in figure 1.

(a) One type I.
(b) One type III. The type III label shall be as shown on figure 5.
(c) One type IV. The information to be shown on the type IV label shall be "USE LAMP NSN6240-00-299-7252 ONLY."

Instruction sheet: An instruction sheet containing the following information shall be furnished with each light. "This light is designed for use with a separately mounted ballast and starter box shown on MIL-F-16377/47, symbol 455. The ballast and starter box is not furnished with the light and it must be ordered separately."

Weight: 3 lbs 13 oz maximum.

QUALITY ASSURANCE: Quality assurance shall be as specified in MIL-F-16377 and as follows:

First article inspection:	
Complete fixture:	See table I
Window assemblies:	See table II
Quality conformance:	
Comparison inspection:	Same as the first article inspection
Inspection of product for delivery:	
Examination:	See MIL-F-16377
Examination for preparation of delivery:	See MIL-F-16377
Quality conformance tests:	See MIL-F-16377

GENERAL INFORMATION:

Assignment of Military part numbers and general information is as follows:

(a) The light described herein is intended for use on water gages.
(b) The light will be attached to the foundation plate by 8 No. 8-32-UNC-2A, $\frac{5}{16}$" long screws provided with the light. The foundation plate is secured to the face of the water gage.
(c) The lens will be retained in place by the foundation plate.
(d) The ballast box and starter shown on MIL-F-16377/47 will not be supplied with light.
(e) When lights are furnished as replacement parts or without mounting plate, window shall be securely taped to fixture to prevent breakage during shipment.
(f) Assignment of Military part numbers and general information is shown in table III.

Source: https://assist.dla.mil -- Downloaded: 20261O-18T08:39Z
Check the source to verify that this is the current version before use.

TABLE I. First article and comparison inspection for complete fixtures.

Tests	Requirement paragraph	Test paragraph	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Enclosure effectiveness: Symbol 154	3.5.11	4.8.14.4	Watertight
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	
Enclosure effectiveness: Symbol 152 Symbol 154	3.5.11	 4.4.14.1 4.4.14.4	 Totally enclosed Watertight
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Continuity of grounding	3.6.13.4	4.8.12	
Optical uniformity	3.7.4.1.1	4.8.4	
Magnetic permeability	3.4.8	4.8.18	
Temperature	3.5.17	4.8.19	

TABLE II. First article and comparison inspection for window assemblies.

Tests	Requirement paragraph	Test paragraph	Remarks
Examination	3.13	4.5	
Optical uniformity	3.7.4.1.1	4.8.4	
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight

TABLE III. Part number and general information.

Military Part number M16377/2-	Item name	National Stock Number (NSN)	Supersedes part number
152	Complete fixture, Symbol 152		MS17240
154	Complete fixture, Symbol 154	6210-00-768-7190	
001	Window assembly	6210-00-676-9397	MS17237-1

Preparing activity:
Navy - SH
(Project 6210-N550)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:39Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY MAIL

FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

COMMANDER
NAVAL SEA SYSTEMS COMMAND (SEA 3112)
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20362

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:39Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-F-16377/2A

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1426 1 OCT 76

Replaces edition of 1 Jan 72 which may be used.

S/N 0102-LF-001-4260

Source: https://assist.dla.mil -- Downloaded: 202610-18T08:39Z
Check the source to verify that this is the current version before use.